

9.1 Preference Data and Reward Models

Humans (or a proxy) pick chosen over rejected. A reward model scores completions.

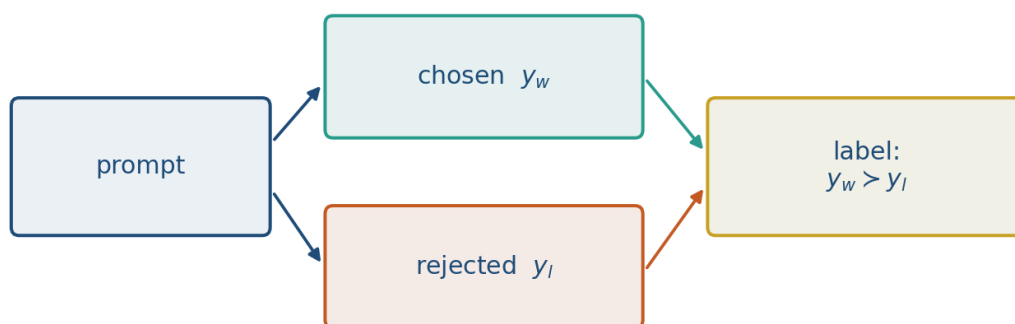
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

SFT copies demonstrations. **Preferences** say which of two replies is better when both are fluent. A **reward model (RM)** is a classifier of that comparison. Week 9.2 will use the RM inside RL; Week 9.3 will skip it.

1. Chosen versus rejected

A human (or a strong teacher model) sees a prompt x and two completions y_w (**chosen**, winner) and y_l (**rejected**, loser). The label is a pairwise order, not a numeric grade. Ties and multi-way ranks exist; this course uses pairs.

One preference row is a comparison, not a score out of 100



Collection is the bottleneck: instructions must be clear, raters disagree, and the pair set can encode rater bias. If both y_w and y_l are wrong, the RM still learns a relative order among mistakes.

2. The reward model

Initialize from the SFT policy (or its encoder). Score a pair with a scalar head $r_\phi(x, y)$. Bradley–Terry / logistic loss:

$$\mathcal{L}_{\text{RM}} = -\log\sigma\left(r_\phi(x, y_w) - r_\phi(x, y_l)\right).$$

The RM wants a higher number on the chosen reply. Lab 9 will fit this on toy scores. At test time you can rank n samples from the policy by r_ϕ (**best-of- n**) without any RL.

A reward model is a classifier of “better completion,” trained on pairs



3. What the number is not

r_ϕ is not truth, safety, or “helpfulness” in the abstract. It is a fit to the raters you hired, on the prompts you wrote. Reward hacking: the policy learns to please r_ϕ with verbose, sycophantic, or empty-safe text. That is why RLHF adds a KL penalty to the SFT model (next note), and why you keep a held-out preference set.

4. Practice

1. If $r(x, y_w) = r(x, y_l)$ always, what is $\mathcal{L}_{\text{RM}}$?
2. Why initialize the RM from SFT instead of from a raw pretrained LM?
3. Best-of- n uses the RM without PPO. What extra cost did you pay at inference?